1 **(a)** $\tan(38°) = \dfrac{TB}{45}$ **[M1]**

35.2 (or 35.15 to 35.16) **[A1]**

(b) $\cos(38°) = \dfrac{45}{PT}$ or $PT^2 = 45^2 + 35.15^2$ **[M1]**

57.1 (or 57.10 to 57.11) **[A1]**

(c) $\tan(XPB) = \dfrac{20}{45}$ **[M1]**

Angle $= \tan^{-1}\left(\dfrac{20}{45}\right)$ **[M1]**

24.0 (or $23.96\ldots$) **[A1]**

[Total Marks: 7]

2 **(a)** Interior angle between parallel North lines $= 180° - 52° = 128°$ **[M1]**

Angles around point $B = 360° - 128° - 142° = 90°$ with no errors seen **[A1]**

(b) $AC = \sqrt{15^2 + 20^2}$ **[M1]**

$AC = \sqrt{625}$ **[M1]**

25 **[A1]**

(c) $\tan(BAC) = \dfrac{20}{15}$ **[M1]**

Angle $BAC = 53.13°$ **[A1]**

Bearing $= 052° + 53.13°$ **[M1]**

105.1 (or $105.13\ldots$) **[A1]**

[Total Marks: 9]

3 **(a)** $AC = \sqrt{8^2 + 15^2}$ **[M1]**

17 **[A1]**

(b) $AD^2 + 12^2 = 17^2$ (or their AC^2) **[M1]**

$AD = \sqrt{289 - 144} = \sqrt{145}$ **[M1]**

12.0 (or $12.04\ldots$) **[A1]**

(c) Area of $\triangle ADC = \dfrac{1}{2} \times 12 \times \sqrt{145} = 72.25\ldots$ **[M1]**

$\dfrac{1}{2} \times 17 \times h = 72.25\ldots$ **[M1]**

8.50 (or $8.499\ldots$) **[A1]**

(d) $\tan(BAC) = \dfrac{15}{8} \implies$ Angle $BAC = 61.93°$ **[M1]**

$\tan(CAD) = \dfrac{12}{\sqrt{145}} \implies$ Angle $CAD = 44.90°$ **[M1]**

Angle $DAB = 61.93° + 44.90°$ **[M1]**

106.8 (or 106.82 to 106.83) **[A1]**

[Total Marks: 12]

4 Identifies cosine ratio: $\cos(\theta) = \dfrac{2.1}{6.5}$ **[M1]**

$\theta = \cos^{-1}\left(\dfrac{2.1}{6.5}\right)$ **[M1]**

71.2 **[A1]**

[Total Marks: 3]

5 (a) Calculates $\angle PAB = 360° - (180° - 40°) - 130° = 90°$ (or similar valid reasoning to show right angle) **[M1]**

Uses Pythagoras' theorem: $PB^2 = 80^2 + 60^2$ **[M1]**

100 **[A1]**

(b) Uses trigonometry to find $\angle APB$: $\tan(APB) = \dfrac{60}{80}$ **[M1]**

$\angle APB = \tan^{-1}(0.75) = 36.869...°$ **[A1]**

Bearing $= 040° + 36.869...° = 076.9$ (accept 077) **[A1]**

(c) Recognises the shortest distance d is the perpendicular height of $\triangle PAB$ from A to PB. **[M1]**

Uses area equation: $\dfrac{1}{2} \times 100 \times d = \dfrac{1}{2} \times 80 \times 60$ OR uses trigonometry $80\sin(53.13°)$ or $60\sin(36.87°)$ **[M1]**

48 **[A1]**

[Total Marks: 9]

6 (a) Sets up equation for $\triangle BCD$: $\tan(42°) = \dfrac{CD}{BC}$

BC $= \dfrac{CD}{\tan(42°)}$ [M1]

Sets up equation for $\triangle ACD$: $\tan(25°) = \dfrac{CD}{BC + 50}$

BC + 50 $= \dfrac{CD}{\tan(25°)}$ [M1]

Equates or substitutes to eliminate BC: $50 = \dfrac{CD}{\tan(25°)} - \dfrac{CD}{\tan(42°)}$ **[M1]**

Factorises to solve for CD: $CD\left(\dfrac{1}{\tan(25°)} - \dfrac{1}{\tan(42°)}\right) = 50$ **[M1]**

48.4 (accept 48.36...) **[A1]**

(b) Recognises the shortest distance d is the perpendicular from B to AD, forming a right-angled triangle with hypotenuse $AB = 50$. **[M1]**

Sets up trigonometric ratio: $\sin(25°) = \frac{d}{50}$ **[M1]**

21.1 (accept 21.13...) **[A1]**

[Total Marks: 8]